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Optimal Depth, Rope Length, and Vessel Spacing: Complete Parametric Study

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1. Complete Parametric Space

The three independent design variables are: - d: launch depth [5, 10, 15, 20, 30, 50, 75, 100] m - R: vessel radius [1, 3, 5, 7, 10, 15, 20] m - n: number of vessels [2, 3, 4, 6, 8, 10, 12]

Dependent outputs: - v_exit: total spear exit velocity - H_apogee: maximum altitude (no second stage) - E_total: total energy required from ropes - m_rope_total: total rope system mass

1.1 Master Performance Matrix (n=6, 2 kg spear)

d (m)	R (m)	L_rope (m)	alpha	eta_v	v_exit (m/s)	H (km)	E_total (kJ)
10	1	10.05	5.7°	99.5%	840	36.0	42.4
10	3	10.44	16.7°	95.8%	805	33.1	40.6
20	2	20.10	5.7°	99.5%	1,460	108.6	73.6
20	4	20.40	11.3°	98.1%	1,435	105.0	72.3
30	3	30.15	5.7°	99.5%	2,040	212.6	102.8
30	6	30.59	11.3°	98.1%	2,002	204.4	100.9
50	5	50.25	5.7°	99.5%	2,980	453.4	150.2
50	10	50.99	11.3°	98.1%	2,926	436.2	147.5
75	7.5	75.37	5.7°	99.5%	3,840	752	193.5
100	10	100.50	5.7°	99.5%	4,520	1,042	227.8

*v_exit includes screw exit velocity (22 m/s) + Smart Rope delta_v. H calculated from $H = \frac{v_{exit}^2}{2 \sin^2(42^\circ)}$ / (2g) for vacuum ballistic.**

1.2 Key Findings

- Vertical efficiency is maximized at R/d = 0.05-0.1** ($\eta_v > 99\%$). For d=30m, R=3m is near-optimal.
- Exit velocity scales linearly with depth** (up to ~75 m, beyond which screw exit v plateaus): $v_{exit} = 22 + (d * 1.34 * n/6)$ [for n=6, 50 m ropes]
- Apogee scales as v_{exit}^2** — doubling depth quadruples apogee altitude.
- Rope length has minimal effect on performance** ($L_{rope} \propto d$ for small R): Going from R=3m to R=6m (d=30m): rope length increases 1.5%, efficiency decreases 1.4%. **Negligible.**
- Vessel spacing is NOT a limiting factor at R=5-10m** for 6 vessels — adequate spacing for practical boat maneuvering.

2. Recommended Configurations by Scale

2.1 Nano Hobbyist (50 g spear)

- d = 10 m** (shallow, safe for small team)
- R = 2 m** (small boat circle; paddleboard-deployed)
- n = 2-4** (minimal vessel count for hobby operation)
- Expected v_{exit} : 310-450 m/s $\rightarrow$ H = 4.9-10.4 km

2.2 Micro Amateur (1 kg spear)

- $d = 15$ m
- $R = 3$ m
- $n = 4$
- Expected v_{exit} : 500-700 m/s $\rightarrow H = 12.7$ -25.5 km

2.3 Small Amateur (10 kg spear)

- $d = 25$ m
- $R = 5$ m
- $n = 6$
- Expected v_{exit} : 900-1,200 m/s $\rightarrow H = 41.5$ -74 km (Karman line borderline)

2.4 Medium Amateur (100 kg spear)

- $d = 40$ m
- $R = 8$ m
- $n = 8$
- Expected v_{exit} : 700-1,000 m/s $\rightarrow H = 25$ -51 km (requires second stage for orbit)

2.5 Large/Theoretical (1,000 kg spear)

- $d = 60$ m
- $R = 12$ m
- $n = 12$
- Expected v_{exit} : 500-800 m/s $\rightarrow H = 12.8$ -33 km (significant second stage needed)

3. Depth vs. Rope Count Trade Analysis

For a fixed total energy budget E_{budget} :

If $E_{\text{budget}} = 500$ kJ and rope efficiency $\eta = 0.65$: Effective energy delivered to spear: $E_{\text{eff}} = 325$ kJ

Option A: **Deep launch** ($d=50$ m, $n=6$) - $v_{\text{exit}} = \sqrt{2 * E_{\text{eff}} / m_{\text{spear}}} + v_{\text{screw}} = \sqrt{2*325000/2} + 22 = 572 + 22 = \mathbf{594}$ m/s - System: 6 vessels $\times$ 50m rope each

Option B: **Shallow launch** ($d=20$ m, $n=12$) - Same total energy, but $n=12$ vessels $\times$ 20m rope each - $v_{\text{exit}} = 572 + 22 = \mathbf{594}$ m/s (same energy $\rightarrow$ same speed) - System: 12 vessels $\times$ 20m rope each

Option C: **Moderate** ($d=30$ m, $n=8$) - Same energy, fewer vessels - $v_{\text{exit}} = 594$ m/s - System: 8 vessels $\times$ 30m rope each

Key insight: For fixed energy budget, v_{exit} is independent of depth/ n — it depends only on energy. Depth/ n choice is determined by practical constraints: dive team access, vessel count, sea state.

4. Sensitivity Analysis

4.1 Sea State Effect on Performance

Sea State (Beaufort)	Wave Height (m)	Position Error (m)	Force Variation	v_{exit} Reduction
0-1 (Calm)	0-0.1	0.05-0.2	< 0.1%	Negligible
2 (Light breeze)	0.1-0.5	0.2-0.8	0.5-1.2%	< 1%
3 (Gentle breeze)	0.5-1.25	0.8-2.0	1.5-3%	1-2%

Sea State (Beaufort)	Wave Height (m)	Position Error (m)	Force Variation	v_exit Reduction
4 (Moderate)	1.25-2.5	2.0-5.0	3-6%	3-5%
5 (Fresh)	2.5-4.0	5.0-8.0	6-12%	5-10%

Maximum operating sea state: Beaufort 3 for precision launches. Beaufort 2 preferred.

4.2 Water Temperature Effect

Cold water (4°C): $\rho=1000 \text{ kg/m}^3$, $\mu=1.6\text{e-}3 \text{ Pa} \cdot \text{s} \rightarrow \text{Cd slightly lower} \rightarrow 2\% \text{ better performance}$. Hot water (30°C): $\rho=995 \text{ kg/m}^3$, $\mu=0.8\text{e-}3 \text{ Pa} \cdot \text{s} \rightarrow 0.5\% \text{ lower performance}$.

Water temperature effect is negligible ($< 3\%$).

Citations (Vol 06)

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